

Chapter 10: Bet Sizing

Advances in Financial Machine Learning (Book 1, Marcos M. López de Prado)

AI Finance Seminar

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Chapter 10 Overview: Why Bet Sizing Matters

The Poker Analogue: Sizing vs. Prediction

- **Cards vs. Prices:** Poker players cannot predict cards with certainty, yet the same top players win year after year because **bet sizing is more critical than raw prediction**.
- **Trading Reality:** High prediction accuracy (60% precision) with 90% recall will still lose money if bet sizes are small on true positives and large on false positives.
- **Decoupling Direction and Size:** Meta-labeling decouples the side prediction (buy/sell) from the size prediction ($m \in [-1, 1]$).

Core Sizing Objectives

- **Turnover Control:** Avoid overtrading on minor price fluctuations while fully exploiting high-confidence signals.
- **Dynamic Adaptation:** Adjust positions as market prices p_t move toward or away from forecasted prices f_i .
- **Risk Budgeting:** Ensure maximum leverage is reached only when multiple concurrent signals confirm market edge.

Topic 10.2: Strategy-Independent Bet Sizing Approaches

1. Bet Concurrency & Mixture of Gaussians (EF3M)

- Calculate net concurrent bets $c_t = c_{t,l} - c_{t,s}$, where $c_{t,l}$ and $c_{t,s}$ are active long/short bet counts.
- Fit a mixture of two Gaussians $F[x]$ on $\{c_t\}$ (López de Prado & Foreman, 2014). Bet size is derived as:

$$m_t = \begin{cases} \frac{F[c_t] - F[0]}{1 - F[0]} & \text{if } c_t \geq 0 \\ \frac{F[c_t] - F[0]}{F[0]} & \text{if } c_t < 0 \end{cases}$$

- The stronger the signal relative to historical concurrency, the smaller the probability of a stronger signal, yielding a larger bet size.

2. Budgeting Approach

- Compute maximum historical concurrent bets $\max_i \{c_{i,l}\}$ and $\max_i \{c_{i,s}\}$.
- Set bet size proportionally: $m_t = \frac{c_{t,l}}{\max_i \{c_{i,l}\}} - \frac{c_{t,s}}{\max_i \{c_{i,s}\}} \in [-1, 1]$. Prevents reaching maximum position before signal concurrency peaks.

Topic 10.3: Bet Sizing from Predicted Probabilities

Binary Classification ($x \in \{-1, 1\}$)

- Test null hypothesis $H_0 : p[x = 1] = \frac{1}{2}$ (no predictive edge).
- Compute test statistic $z \sim \mathcal{N}(0, 1)$:

$$z = \frac{p[x = 1] - \frac{1}{2}}{\sqrt{p[x = 1](1 - p[x = 1])}} \in (-\infty, +\infty)$$

- Derive continuous bet size $m \in [-1, 1]$ using standard Normal CDF $Z[\cdot]$:

$$m = 2Z[z] - 1$$

Multi-Class One-vs-Rest ($x \in X, \|X\| > 2$)

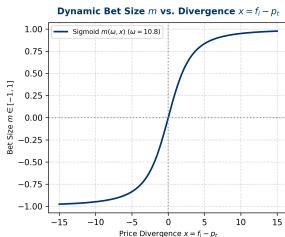
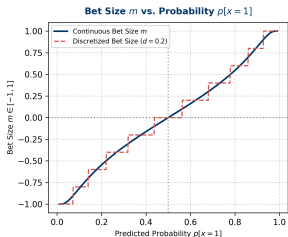
- Let $\tilde{p} = \max_i \{p_i\}$ be maximum predicted probability across $\|X\|$ classes. Test $H_0 : \tilde{p} = \frac{1}{\|X\|}$.
- Compute z-statistic:

$$z = \frac{\tilde{p} - \frac{1}{\|X\|}}{\sqrt{\tilde{p}(1 - \tilde{p})}} \in [0, +\infty)$$

- Bet size includes side implied by predicted class x :

$$m = x \cdot (2Z[z] - 1) \in [-1, 1]$$

Topic 10.3 (Cont.): Mapping Probabilities to Bet Sizes



Key Graphical Insights

- **S-Curve Shape:** Smooth non-linear transition from $m = -1$ ($p \rightarrow 0$) to $m = +1$ ($p \rightarrow 1$).
- **Zero Bet at Neutral Edge:** When $p = 0.5$, $z = 0 \implies m = 2(0.5) - 1 = 0$ (no position taken).
- **Rapid Scale-Up:** Position size increases rapidly once probability moves beyond noise threshold ($p > 0.6$ or $p < 0.4$).
- **Discretization Overlay:** Discretizing with step size $d = 0.2$ creates a step function, filtering out micro-adjustment trade noise.

Topic 10.4 & 10.5: Averaging Active Bets & Size Discretization

Averaging Concurrent Active Bets

- Each bet spans from origin time $t_{i,0}$ to barrier exit time $t_{i,1}$ (Triple-Barrier Method).
- Overriding old bets causes excessive turnover.
- **Solution:** Average signals across all active bets open at time t :

$$\bar{m}_t = \frac{1}{\|Active_t\|} \sum_{i \in Active_t} m_{i,t}$$

- Smooths out temporary noise and reduces unnecessary rebalancing trades.

Size Discretization ($d \in (0, 1]$)

- Continuous bet size m generates small jitter trades on minor probability shifts.

- **Discretization Formula:**

$$m^* = \text{round} \left[\frac{m}{d} \right] \times d$$

- Capped within range $[-1, 1]$.
- **Practical Impact:** $d = 0.1$ or 0.2 enforces minimum execution thresholds (e.g., 10% or 20% portfolio changes), cutting transaction costs.

Topic 10.6: Dynamic Bet Sizes & Price Divergence

Modelling Position Size from Price Divergence $x = f_i - p_t$

- As market price p_t moves relative to forecasted price f_i , price divergence $x = f_i - p_t$ changes dynamically.
- Sigmoid Sizing Function:** Bounded target position $\hat{q}_{i,t} \in (-Q, Q)$ for max position Q :

$$\hat{q}_{i,t} = \text{int} [m[\omega, f_i - p_t] \cdot Q], \quad \text{where } m[\omega, x] = \frac{x}{\sqrt{\omega + x^2}}$$

- Calibration of ω :** Calibrated via user-defined pair (x^*, m^*) (e.g., $m^* = 0.95$ when divergence $x^* = 10$):

$$\omega = x^{*2} (m^{*-2} - 1)$$

- As $p_t \rightarrow f_i$, divergence $x \rightarrow 0 \implies \hat{q}_{i,t} \rightarrow 0$, naturally realizing profits as price target is reached!

Topic 10.6 (Cont.): Breakeven Limit Price & Order Execution

Inverse Price Function $L[f_i, \omega, m]$

- Inverse function of $m[\omega, f_i - p_t]$ with respect to p_t :

$$L[f_i, \omega, m] = f_i - m \sqrt{\frac{\omega}{1 - m^2}}$$

- Monotonic function ensuring the algorithm never realizes losses as $p_t \rightarrow f_i$.

Breakeven Limit Price $\bar{p}$

- For required order size $\hat{q}_{i,t} - q_t$, compute average limit price $\bar{p}$ across position steps j :

$$\bar{p} = \frac{1}{|\hat{q}_{i,t} - q_t|} \sum_{j=|q_t + \text{sgn}(\hat{q}_{i,t} - q_t)|}^{|\hat{q}_{i,t}|} L\left[f_i, \omega, \frac{j}{Q}\right]$$

- Ensures execution orders are placed strictly between current price p_t and target f_i without realizing loss.

Summary: Key Takeaways on Bet Sizing

Core Principles for Quantitative Asset Managers

- 1 **Bet Sizing Equals Alpha Preservation:** Price prediction alone cannot guarantee profitability; proper bet sizing prevents bad trades on false positives from ruining portfolio PnL.
- 2 **Meta-Labeling Integration:** Use primary models for trade direction (Side) and secondary ML probabilities for bet size ($m \in [-1, 1]$).
- 3 **Tame Turnover via Averaging & Discretization:** Average concurrent active bets over holding periods $[t_{i,0}, t_{i,1}]$ and discretize sizes ($m^* = \text{round}(m/d)d$) to eliminate jitter trades.
- 4 **Dynamic Limit Pricing:** Scale positions based on price divergence $x = f_i - p_t$ using calibrated Sigmoid functions, enforcing breakeven limit prices $\bar{p}$ to lock in gains.

Thank You!

Questions & Discussion

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Reference: Marcos M. López de Prado, *Advances in Financial Machine Learning* (Chapter 10)